

VARUN SATISH

☎ +1 (447) 902-1825 ✉ satish5@illinois.edu [in linkedin.com/in/varunsatish](https://www.linkedin.com/in/varunsatish) github.com/varunsatish14

Education

University of Illinois

Bachelor of Science in Astrophysics, Astronomy + Data Science. GPA: 3.57

Aug. 2022 – Present

Urbana, IL

The International School, Bangalore

International Baccalaureate Diploma Program

Aug. 2018 – May 2022

Bangalore, IND

Relevant Coursework

- Computing in Astronomy
- Observational Astronomy
- Planetary Systems
- Differential Equations
- Data Curation
- Data Science
- Galaxies
- Linear Algebra

Research Experience / Projects

Cosmology/Radiative Transfer Research | Kirk Barrow Research Group

August 2023 – Present

- Conducted research on galaxy evolution and the propagation of radiation from stars in early galactic environments, contributing to a deeper understanding of radiative transfer.
- Utilized Python packages including Numpy, Healpy (HEALPix for Python), yt, and matplotlib, along with parallel computing techniques to process and analyze large simulations.
- Currently investigating ionizing radiation attenuation across galaxies by calculating the Escape Fraction of Ionizing Radiation, a key metric in understanding radiation's journey from stars through the interstellar medium.

Positions

Course Assistant and Grader

Department of Astronomy

August 2023 – Present

Urbana, IL

- Supported night and solar observing sessions for 100-level astronomy classes, guiding students in telescope use at the Campus Observatory.
- Graded and provided feedback on discussion posts for ASTR 330: Extraterrestrial Life with Prof. Dunne, helping foster deeper discussions on astrobiology and extraterrestrial life.
- Providing grading support for ASTR 100: Introduction to Astronomy, ensuring timely feedback on foundational astronomy concepts to help students get introduced to the subject

Technical Skills

Languages: Python, Java, HTML, SQL, $\text{\LaTeX}$

Astronomy Packages: yt, Healpy, Matplotlib, Numpy, MPI, SciPy, astropy

Technologies: Unix, Git, GitHub, VSCode, High Performance Computing, Parallel Computing

Leadership / Extracurricular

Astronomical Society at the University of Illinois

President

April 2024 – Present

- Achieved record-breaking recruitment, growing society membership to over 300 members within the first year of presidency.
- Collaborated with the Astronomy Department to attract and engage prospective students by representing the society at admissions events and orientations.
- Led the society in planning and executing Observatory Open Houses, new student orientation activities, and community star parties, fostering a vibrant community of astronomy enthusiasts.

Illinois Space Society

Educational Outreach Board

August 2022 – May 2024

- Joined the Educational Outreach Board to contribute to initiatives that connect local schools with space and astronomy resources.
- Coordinated outreach events with middle and high schools, promoting STEM engagement through inspiring activities.
- Designed and delivered educational content to spark interest in space and astronomy among students, fostering early interest in STEM careers.

- Appointed as an Astronomical Society representative to the Observatory’s board, tasked with promoting preservation and community engagement
- Facilitated cross-collaboration among the Department, Friends of the Observatory, and Society members to align preservation goals for the historic Campus Observatory.
- Spearheaded initiatives to enhance outreach, connecting with alumni and increasing community involvement in observatory activities and history.

Conference Participation

ISRPA STEM Career Exploration and Symposium

July 2024

University of Illinois

- Presented a poster with results on research work conducted during IDEAS 2024
- Gave a presentation to fellow students within the program on my work on the escape fractions of ionizing radiation

Astrofest 2024

April 2024

University of Illinois

- Presented a year-end research poster on computational cosmology under Prof. Kirk Barrow to the Illinois Astronomy community, gaining insights and feedback from peers and professionals.

Awards and Honors

Phi Eta Sigma Honor Society

Since August 2023

Member

Silver Honour

July 2022

The International Astronomy and Astrophysics Competition

Youth (18+) Category

Silver Honour

July 2021

The International Astronomy and Astrophysics Competition

Junior (Under 18) Category

Distinction

July 2020

Cambridge ICE

Certifications

Amateur Radio Extra Class License

October 2023 – Present

Federal Communications Commission

Data Science DISCOVERY

May 2024

University of Illinois

[Link](#)

Accelerated Computer Science Fundamentals

July 2023

University of Illinois

[Link](#)

Data-Driven Astronomy

May 2023

University of Sydney

[Link](#)

16.00x: Introduction to Aerospace Engineering

November 2021

MITx

[Link](#)

Astrophysics XSeries

August 2021

Australian National University

[Link](#)

Understanding Einstein: The Special Theory of Relativity

July 2021

Stanford University

[Link](#)

Astronomy: Exploring Time and Space

May 2021

University of Arizona

[Link](#)

Languages

- English (Native)
- Kannada (Native)
- German (Intermediate)
- Spanish (Intermediate)